

School of Computer Science and Engineering

Experiment List for Programming Ability and Logic Building - 1

Proposed Date	Lecture	Experiment	In Class / Take Home
Week : (19/01/26 to 25/01/26)	1	You are given an array of integers arr[]. You have to reverse the given array. Note: Modify the array in place. Examples: Input: arr = [1, 4, 3, 2, 6, 5] Output: [5, 6, 2, 3, 4, 1] Explanation: The elements of the array are [1, 4, 3, 2, 6, 5]. After reversing the array, the first element goes to the last position, the second element goes to the second last position and so on. Hence, the answer is [5, 6, 2, 3, 4, 1]. Link : https://www.geeksforgeeks.org/problems/reverse-an-array/1	In Class
	1	Given an array arr[]. Your task is to find the minimum and maximum elements in the array. Examples: Input: arr[] = [1, 4, 3, 5, 8, 6] Output: [1, 8] Explanation: minimum and maximum elements of array are 1 and 8. Link: https://www.geeksforgeeks.org/problems/find-minimum-and-maximum-element-in-an-array4428/1	In Class
	1	Given an integer array arr[] and an integer k, your task is to find and return the kth smallest element in the given array. Note: The kth smallest element is determined based on the sorted order of the array. Examples : Input: arr[] = [10, 5, 4, 3, 48, 6, 2, 33, 53, 10], k = 4 Output: 5 Explanation: 4th smallest element in the given array is 5. Link: practice.geeksforgeeks.org/problems/kth-smallest-element/0	Take Home

	1	You are given two arrays a[] and b[], return the Union of both the arrays in any order. The Union of two arrays is a collection of all distinct elements present in either of the arrays. If an element appears more than once in one or both arrays, it should be included only once in the result. Note: Elements of a[] and b[] are not necessarily distinct. Note that, You can return the Union in any order but the driver code will print the result in sorted order only. Examples: Input: a[] = [1, 2, 3, 2, 1], b[] = [3, 2, 2, 3, 3, 2] Output: [1, 2, 3] Explanation: Union set of both the arrays will be 1, 2 and 3. Link: https://www.geeksforgeeks.org/problems/union-of-two-arrays3538/1	Take Home
	1	Given an array arr[]. The task is to find the largest element and return it. Examples: Input: arr[] = [1, 8, 7, 56, 90]	Take Home

School of Computer Science and Engineering

Experiment List for Programming Ability and Logic Building - 1

Proposed Date	Lecture	Experiment	In Class / Take Home
		Output: 90 Explanation: The largest element of the given array is 90. Link: https://www.geeksforgeeks.org/problems/largest-element-in-array4009/0?utm_source=youtube&utm_medium=collab_striver_ytdescription&utm_campaign=largest-element-in-array	
Week : (19/01/2026 to 25/01/2026)	2	Given an array arr, rotate the array by one position in clockwise direction. Examples: Input: arr[] = [1, 2, 3, 4, 5] Output: [5, 1, 2, 3, 4] Explanation: If we rotate arr by one position in clockwise 5 come to the front and remaining those are shifted to the end. Link: https://www.geeksforgeeks.org/problems/cyclically-rotate-an-array-by-one2614/1	In Class

01/ 26)	2	You are given an integer array arr[]. You need to find the maximum sum of a subarray (containing at least one element) in the array arr[]. Note : A subarray is a continuous part of an array. Examples: Input: arr[] = [2, 3, -8, 7, -1, 2, 3] Output: 11 Explanation: The subarray [7, -1, 2, 3] has the largest sum 11. Link: practice.geeksforgeeks.org/problems/kadanes-algorithm/0	In Class
	2	Given a sorted array of distinct integers and a target value, return the index if the target is found. If not, return the index where it would be if it were inserted in order. You must write an algorithm with O(log n) runtime complexity. Example 1: Input: nums = [1,3,5,6], target = 5 Output: 2 Link: https://leetcode.com/problems/search-insert-position/description/?envType=problem-list-v2&envId=array	Take Home
	2	Given an array of integers nums and an integer target, return <i>indices of the two numbers such that they add up to target</i>. You may assume that each input would have exactly one solution, and you may not use the <i>same</i> element twice. You can return the answer in any order. Example 1: Input: nums = [2,7,11,15], target = 9 Output: [0,1] Explanation: Because nums[0] + nums[1] == 9, we return [0, 1]. Link: https://leetcode.com/problems/two-sum/description/?envType=problem-list-v2&envId=array	Take Home

School of Computer Science and Engineering

Experiment List for Programming Ability and Logic Building - 1

Proposed Date	Le ct ur e	Experiment	In Class / Take Home

2	You are given an array arr[] of non-negative numbers. Each number tells you the maximum number of steps you can jump forward from that position. For example: • If arr[i] = 3, you can jump to index i + 1, i + 2, or i + 3 from position i. • If arr[i] = 0, you cannot jump forward from that position. Your task is to find the minimum number of jumps needed to move from the first position in the array to the last position. Note: Return -1 if you can't reach the end of the array. Examples : Input: arr[] = [1, 3, 5, 8, 9, 2, 6, 7, 6, 8, 9] Output: 3 Explanation: First jump from 1st element to 2nd element with value 3. From here we jump to 5th element with value 9, and from here we will jump to the last. Link: https://www.geeksforgeeks.org/problems/minimum-number-of-jumps-1587115620/1	Take Home
---	--	-----------